FUEL DELIVERY MODULE (FDM) HOSE REPLACEMENT

AFFECTED VEHICLES: CERTAIN 2000 MODEL YEAR LINCOLN LS

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OVERVIEW

This procedure is for replacing the rubber fuel hose that is part of the FDM. Performing this procedure requires draining the right-hand side of the fuel tank, through the outlet fitting of the FDM. The FDM is accessed through the floor pan under the rear seat cushion. The fuel tank must be lowered one inch to provide clearance for removal of the FDM. Once the FDM is removed from the tank a ***new*** rubber hose will be installed with ***new*** worm drive clamps.

FUEL DRAINING

WARNING: DO NOT SMOKE OR CARRY LIGHTED TOBACCO OR OPEN FLAME OF ANY TYPE WHEN WORKING ON OR NEAR ANY FUEL-RELATED COMPONENT. HIGHLY FLAMMABLE MIXTURES ARE ALWAYS PRESENT AND CAN BE IGNITED, RESULTING IN PERSONAL INJURY.

1. **WARNING: ADEQUATE VENTILATION IS REQUIRED WHEN WORKING IN A CONFINED AREA WITH FUEL AND FUEL-RELATED COMPONENTS.**

   Lower the vehicle windows and open the moonroof (if equipped) to provide ventilation while servicing the fuel system.

2. Remove the rear seat cushion.
3. Roll back the floor pan insulation padding to access the FDM cover.

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4. Remove the FDM cover. See Figure 1.
5. Disconnect the electrical connector on the FDM. See Figure 1.
6. Install a memory saver, then disconnect the battery negative cable.
7. **WARNING: FUEL IN THE FUEL SYSTEM REMAINS UNDER HIGH PRESSURE EVEN WHEN THE ENGINE IS NOT RUNNING. BEFORE REPAIRING OR DISCONNECTING ANY OF THE FUEL SYSTEM COMPONENTS, THE FUEL SYSTEM PRESSURE MUST BE RELIEVED TO PREVENT ACCIDENTAL SPRAYING OF FUEL, CAUSING PERSONAL INJURY OR A FIRE HAZARD.**

   Release the fuel system pressure as follows:
   - Remove the engine cover to access the Schrader valve mounted on the engine fuel rail.
   - Remove the Schrader valve cap and attach an EFI/CFI fuel pressure gauge.
   - Slowly open the manual valve on the fuel pressure gauge to relieve fuel system pressure.
   - Remove the EFI/CFI fuel pressure gauge and install the cap on the Schrader valve.
   - Install the engine cover.
8. **CAUTION: Place shop towels on the carpeting and trim near the access hole for the FDM to prevent damage from splashed fuel while servicing the fuel system.**

   CAUTION: To disconnect the fuel line connector from the outlet fitting on the module, carefully press down on the fuel line connector while pressing the release tabs. Pull straight up to remove. Failure to follow this procedure could result in damage to the fitting or fuel line.

   Press the release tabs on the FDM outlet fitting, turn the fuel line slightly, and pull straight up to remove the black fuel line connector elbow. See Figure 1.

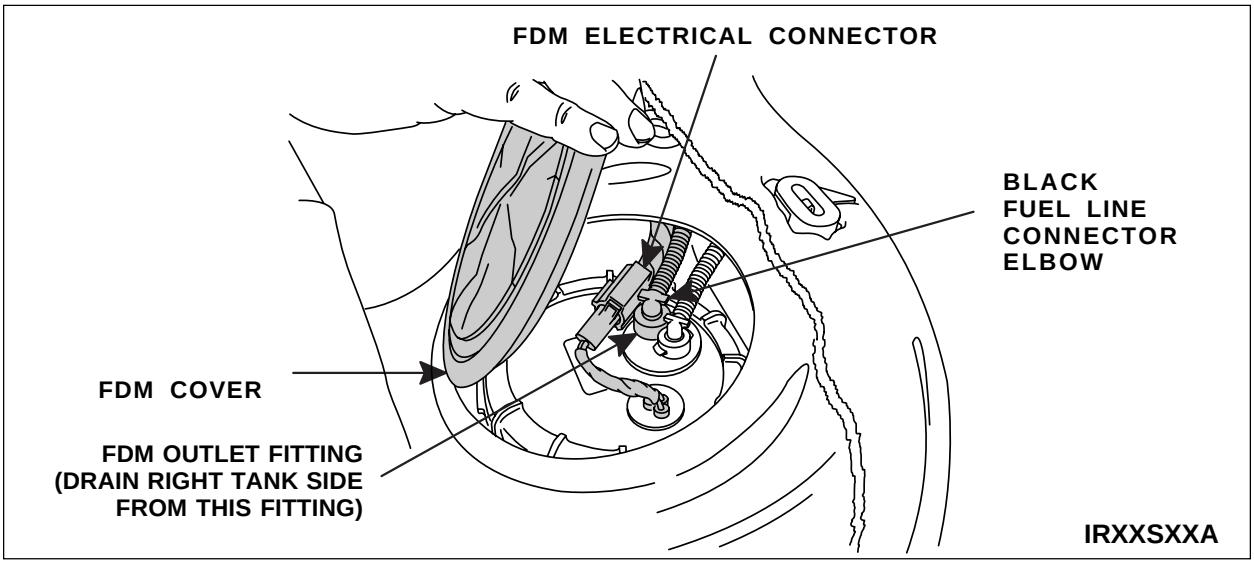


FIGURE 1

9. NOTE: The fuel line draining connector tool (sent to your dealer in a separate mailing must be adapted to the fuel storage tanker hose to drain the fuel.

   Attach the fuel line draining connector tool to the fuel storage tanker hose and the outlet fitting on the FDM. Siphon the fuel until the tank side is empty.

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FDM — REMOVAL

NOTE: The fuel tank must be lowered 25 mm (1 inch) to provide clearance during the FDM removal.

1. Raise and support the vehicle.
2. Remove the center floor pan crossmember. See Figure 2.
3. Remove the retainers from the exhaust heat shield, then lower the heat shield onto the exhaust. See Figure 2.

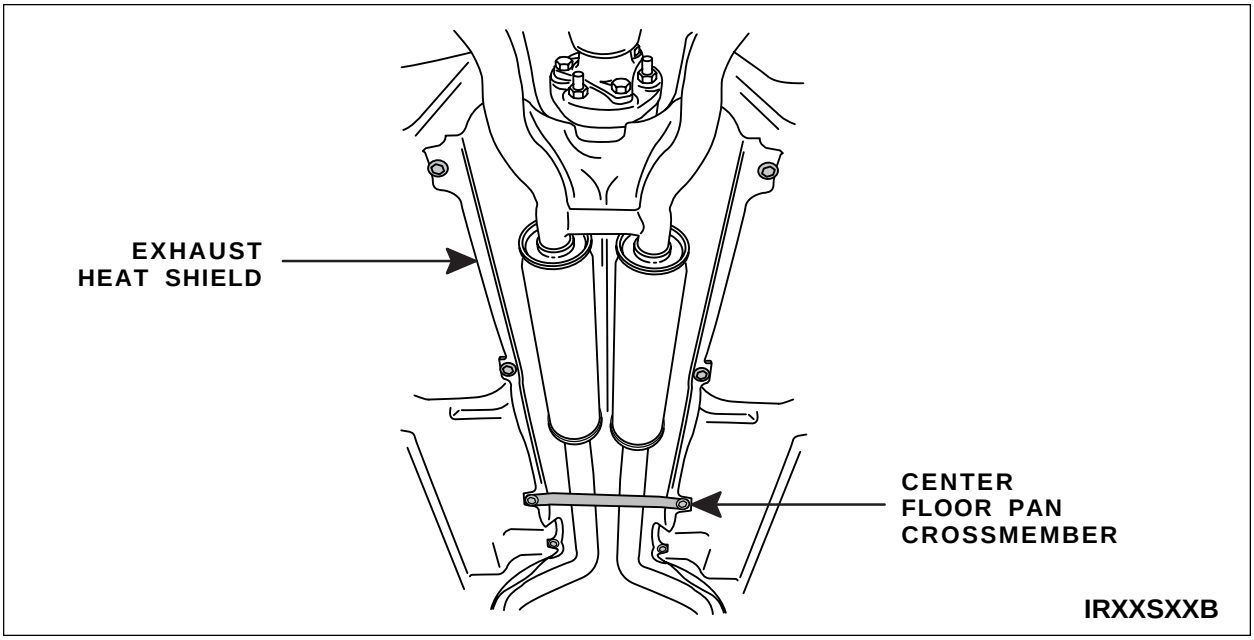


FIGURE 2

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4. Lower the fuel tank by loosening the fuel tank support strap bolts 25 mm (1 inch). See Figure 3.

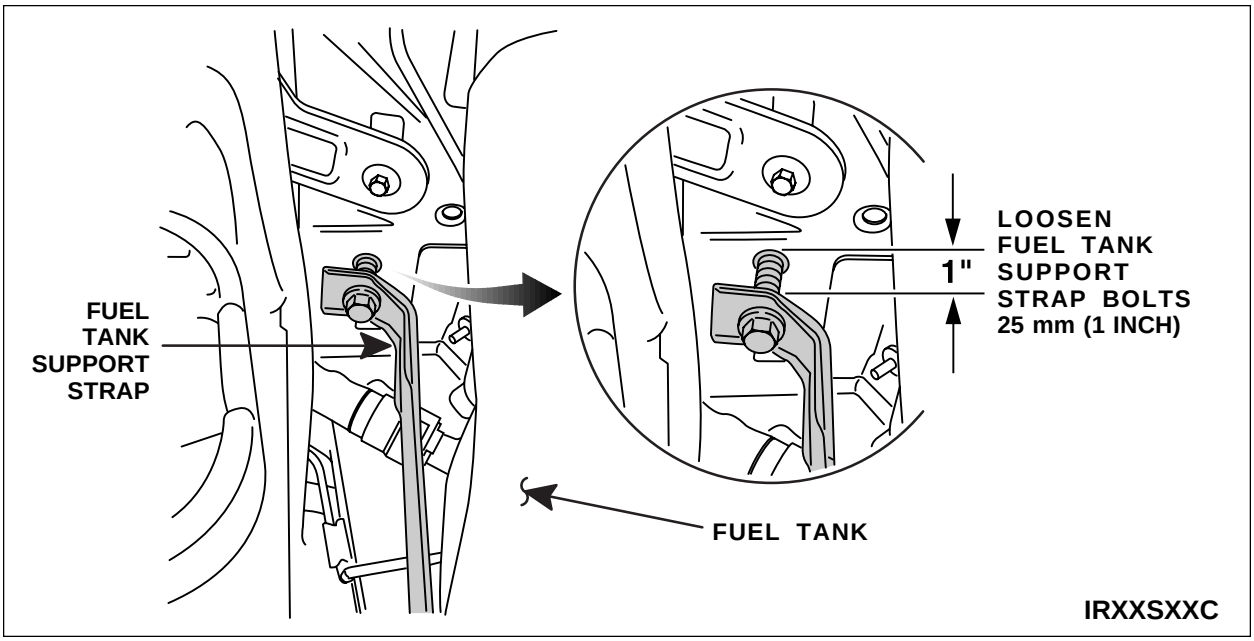


FIGURE 3

5. Lower the vehicle.
6. Press the release tabs on the FDM fitting, turn the fuel line slightly, and pull straight up to remove the white fuel line connector elbow.
7. Remove the FDM lockring retainer nut, using Fuel Sender Wrench 310-069. See Figure 4.

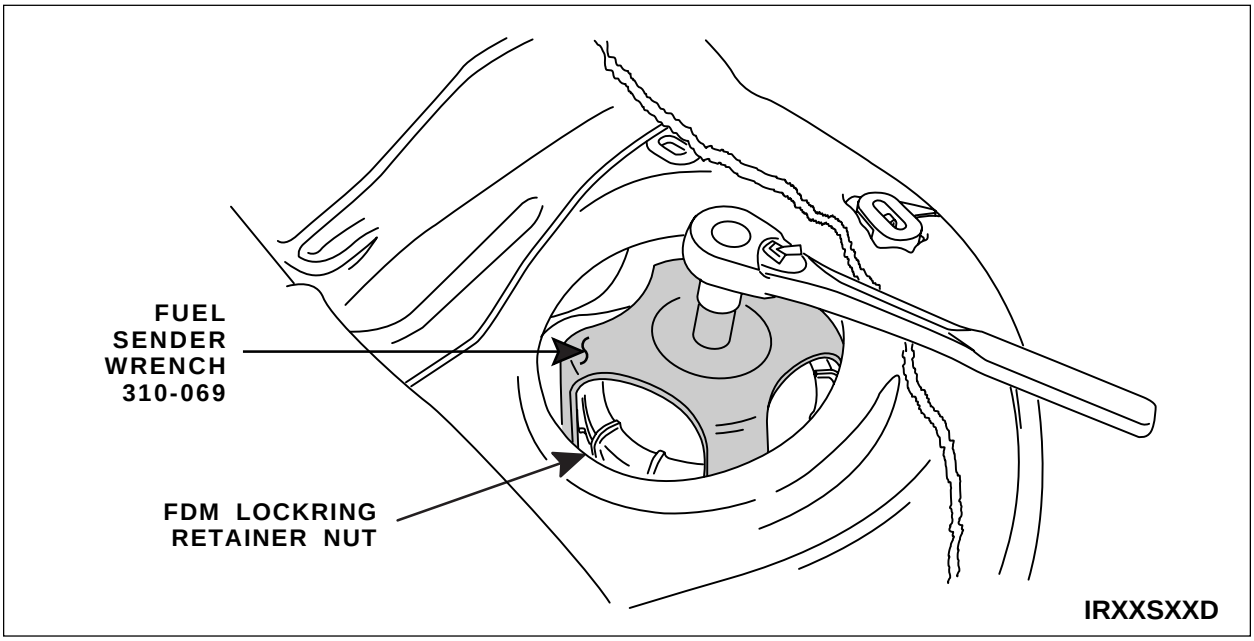
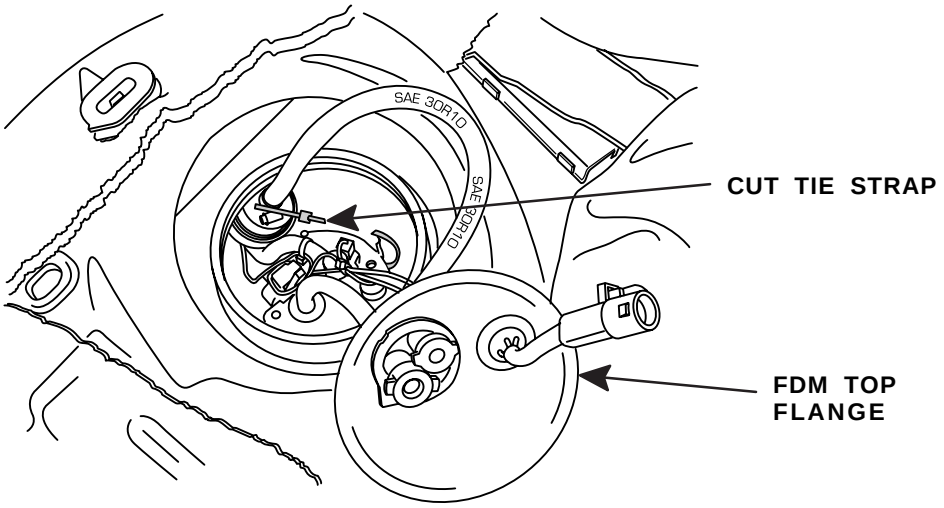


FIGURE 4

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8. Lift up on the FDM top flange and position the flange clear of the access hole.
9. NOTE: The tie strap securing the in-tank filter must be cut prior to removing the FDM from the tank.

   Cut the tie strap securing the in-tank filter to the locking tab of the FDM reservoir. See Figure 5.



TIE STRAP MUST BE CUT BEFORE REMOVING FDM FROM TANK

IRXXSXXE

FIGURE 5

10. **CAUTION: The FDM must be handled carefully to avoid damage to the float arm and the filter during removal.**

    CAUTION: The module will contain some residual fuel. To minimize the possibility of fuel spillage inside the vehicle, place a clean drain pan near the FDM access hole.

    Release the FDM reservoir from the tank-mounting flange by squeezing the locking tabs together. Lift the filter first, then lift the FDM up out of the tank and place into a clean drain pan. Remove the drain pan containing the FDM from the vehicle.

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FUEL HOSE REPLACEMENT

1. Drain the residual fuel from the FDM.
2. NOTE: The hose can be identified by the markings “SAE 30R10”.

   Remove the rubber hose and worm drive clamps between the FDM top flange and the metal in-tank filter. Discard the hose and worm drive clamps, then remove any remaining rubber material from these surfaces. See Figure 6.

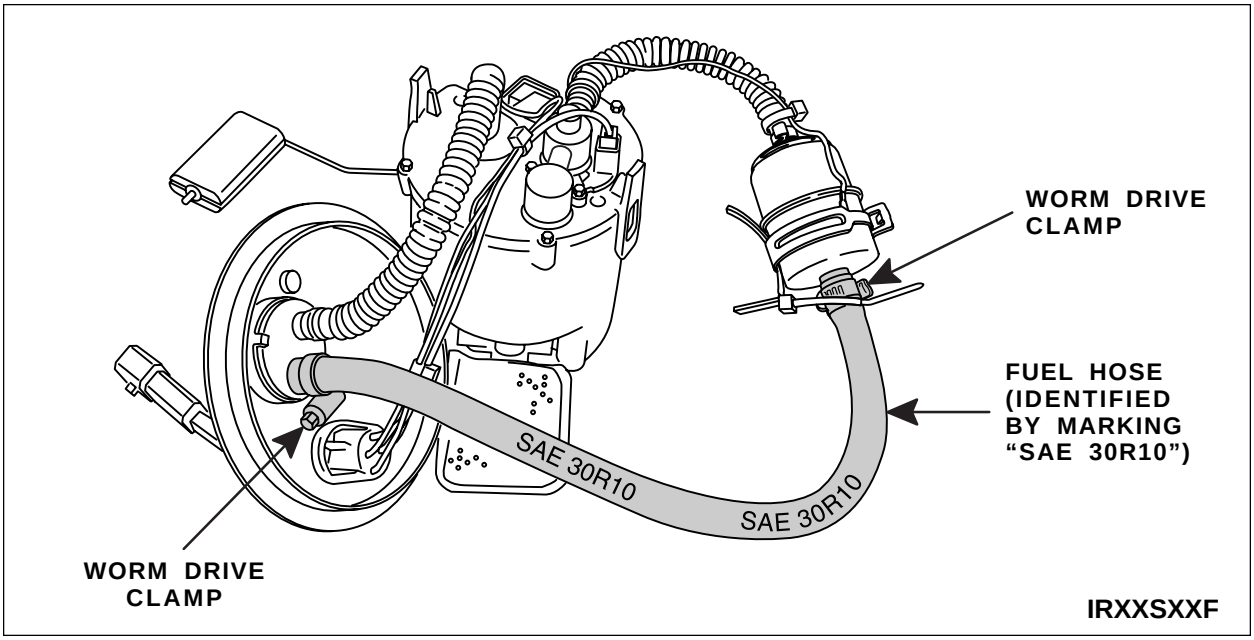


FIGURE 6

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3. Install the ***new*** rubber hose between the FDM top flange and the metal in-tank filter as follows. See Figure 7.
   - Connect the ***new*** rubber hose onto the fitting of the FDM top flange, leaving a 1-2 mm gap between the end of the hose and the surface of the flange.
   - Slide the ***new*** worm drive clamps onto the free end of the hose with the screw heads pointing in a clockwise direction.
   - Connect the free end of the hose onto the metal in-tank filter, leaving a 1-2 mm gap between the end of the hose and the surface of the filter.
4. NOTE: Clamps should be oriented away from other hoses and plastic parts of the FDM.

   Position the ***new*** worm drive clamps so there is a 1-2 mm gap between the end of the hose and the edge of the clamp. Tighten clamps to 2 Nm (18 lb-in). See Figure 7.

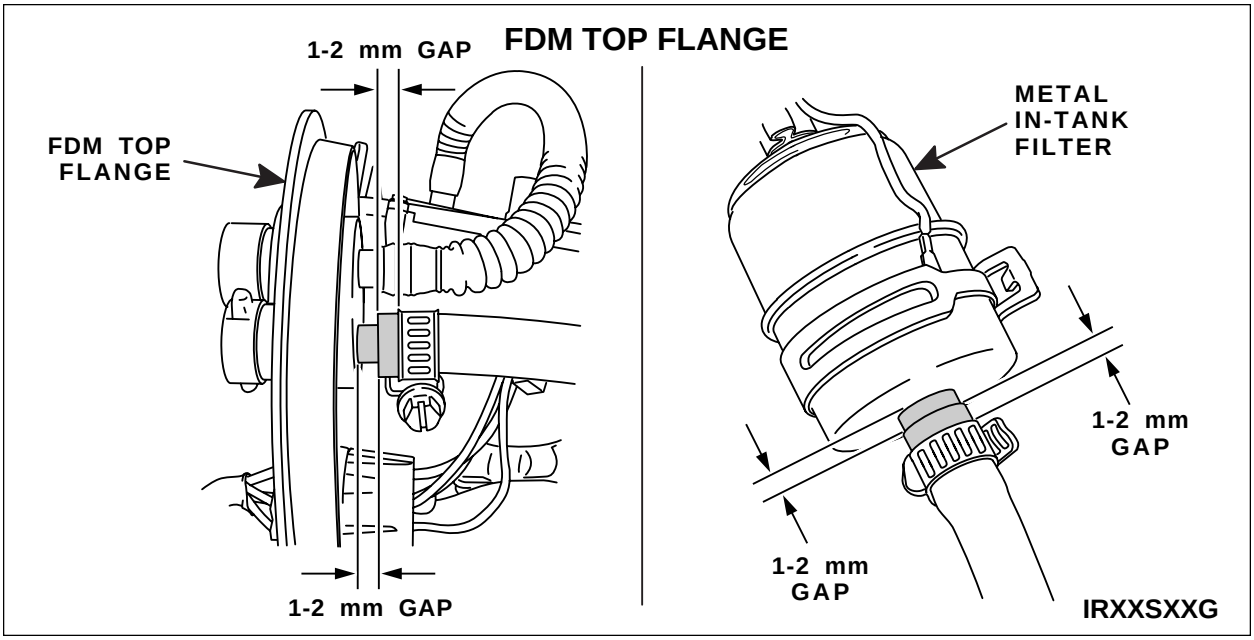


FIGURE 7

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FDM — INSTALLATION

1. Place a paint mark on the metal in-tank filter to show the location of the tie strap. See Figure 8.
2. With the help of an assistant, replace the tie strap on the metal in-tank filter as follows. See Figure 8.
   - Squeeze the constant tension clamp securing the tie strap and ground strap to the filter, then remove the old tie strap.
   - Install a ***new*** tie strap under the clamp. Align the ***new*** tie strap with the paint mark, placing the end of the tie strap 25 mm (1 inch) from the edge of the clamp.
   - Release the constant tension clamp. Verify that the ground strap is still secure under the clamp.
3. Route the ***new*** tie strap through the locking tab on the opposite side of the FDM reservoir from the float.
4. NOTE: Do not tighten the tie strap securing the metal in-tank filter until after the FDM is positioned into the tank.

   Connect the tie strap to itself to form a loop. See Figure 8.

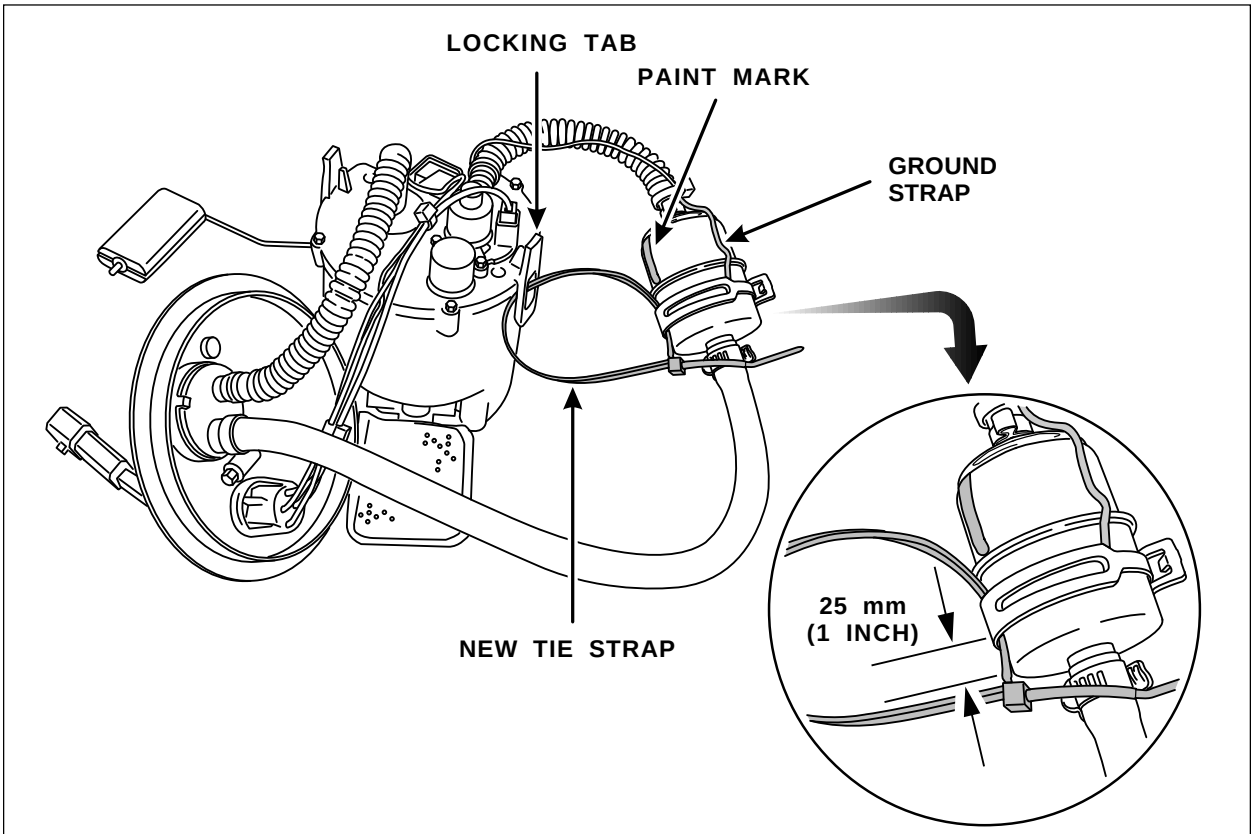


FIGURE 8

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5. Remove the old gasket, then install a ***new*** gasket on the FDM.
6. NOTE: The float of the FDM is positioned toward the centerline of the vehicle.

   Position the FDM into the fuel tank. Engage the locking tabs of the FDM reservoir to the tank-mounting cup until both “click” into place.
7. **CAUTION: To avoid damaging the locking tab, carefully guide the in-tank filter toward the FDM reservoir while tightening the tie strap.**
8. Tighten the tie strap securing the in-tank filter to the locking tab of the FDM. Do not trim the excess tie strap. See Figure 9.

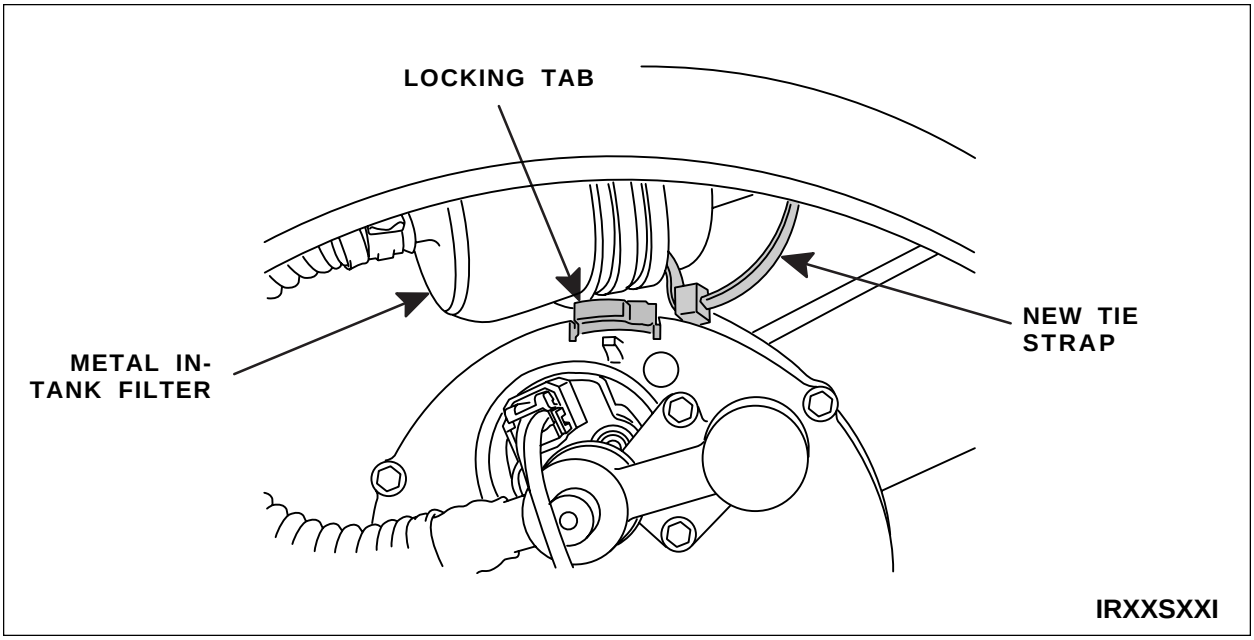


FIGURE 9

9. NOTE: There are two (2) locating tabs on the FDM top flange which engage in matching notches in the opening of the fuel tank.

   Position the FDM top flange onto the tank, then install the lockring retainer nut by hand. Using Fuel Sender Wrench 310-069, tighten the lockring retainer nut to 80 Nm (59 lb-ft).
10. Connect the two (2) fuel line connector elbows to the FDM. Check engagement of the fuel line connector elbows by pulling up on them.
11. Connect the FDM electrical connector.
12. Install the FDM access cover.
13. Remove shop cloths, then reposition the floor pan insulation padding.
14. Install the rear seat cushion.
15. Raise and support the vehicle.
16. Raise the fuel tank by alternately tightening the fuel tank support strap bolts. Final tighten bolts to 35 Nm (26 lb-ft).

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17. Install the exhaust heat shield and center floor pan crossmember. Tighten retainers to 10 Nm (89 lb-in).
18. Lower the vehicle.
19. Connect the battery negative cable, then remove the memory saver.
20. Raise the vehicle windows and close the moonroof (if equipped).
21. Refill the fuel tank.

PREPARING AUTHORIZED MODIFICATIONS LABEL

1. Obtain and prepare the Authorized Modifications Label. Write the modification completed, your P&A Code, date of repair and the recall program type and program number. See Figure 10.

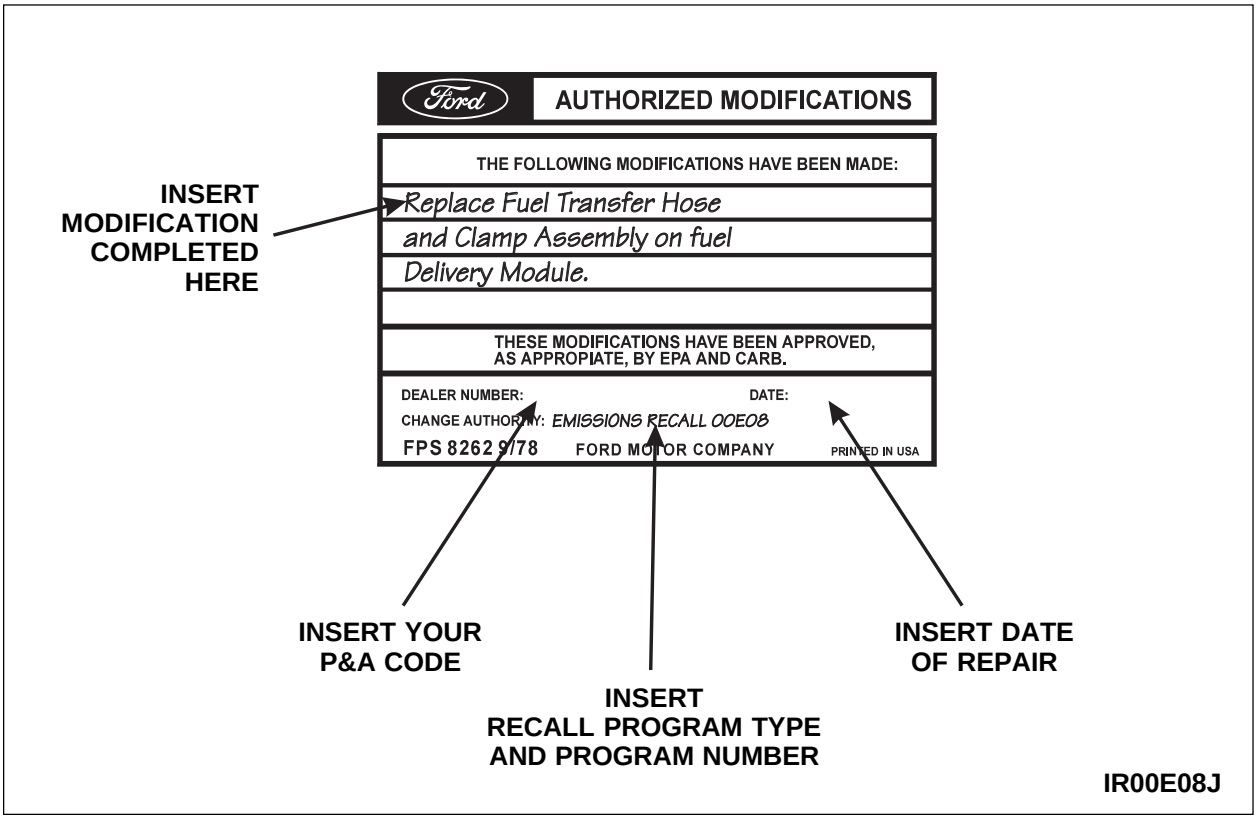


FIGURE 10

2. Apply the label to a clear area of the upper radiator support.
3. Provide California and Massachusetts owners with “Emissions Recall-Proof of Correction” Certificates.

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